

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

*I K. Arghitha (192312527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *K. Arghitha*

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Depth-First Search (DFS).

Using DFS from S, and taking the left child first, the traversal is:

$S \rightarrow A \rightarrow C \rightarrow G \rightarrow H \rightarrow D \rightarrow B \rightarrow E \rightarrow I \rightarrow F$

For the given 5-row table:

Iteration	Stack Before	Expanded	Stack After
1.	[S]	S	[B, A]
2.	[B, A]	A	[B, D, C]
3.	[B, D, C]	C	[B, D, H, G]
4.	[B, D, H, G]	G	[B, D, H]
5.	[B, D, H]	H	[B, D]

DFS order:

$S \rightarrow A \rightarrow C \rightarrow G \rightarrow H \rightarrow D \rightarrow B \rightarrow E \rightarrow I \rightarrow F$

In the stack, the rightmost node is expanded first because DFS uses LIFO.

Uniform Cost Search (UCS)

Start node = A

Goal node = D

Edge cost:

- $A \rightarrow B = 2$
- $A \rightarrow E = 4$
- $B \rightarrow E = 1$
- $B \rightarrow D = 5$
- $E \rightarrow D = 3$
- $E \rightarrow C = 2$

Priority Queue Trace.

Step	Priority queue.	Expanded	cost.
1.	[A(0)]	A	0
2	[B(2), E(4)]	B	2
3	[E(3), D(7)]	E	3
4	[C(5), D(6)]	C	5
5	[D(6)]	D	6

Explanation

Step 1: Start at A with costs 0

From A:

$$\bullet B = 0 + 2 = 2$$

$$\bullet E = 0 + 4 = 4$$

So B is selected next.

Step-2: Expand B with cost 2.

From B:

$$\bullet E = 2 + 1 = 3$$

$$\bullet D = 2 + 5 = 7$$

E with cost 3 is Selected.

Step-3: Expand E with cost 3

From E:

$$\bullet D = 3 + 3 = 6$$

$$\bullet C = 3 + 2 = 5$$

C is selected next because its cost is 5.

Step 4: Expand c with cost 5. No better path to D is found.

Step 5: D has the lowest cost, 6, so D is reached.

Least-cost path

$A \rightarrow B \rightarrow E \rightarrow D$

$$2 + 1 + 3 = 6$$

Final Answer: Least-cost path is $A \rightarrow B \rightarrow E \rightarrow D$ with initial cost 6.

Problem Formulation

Problem

An AI system must find a route from college Gate to CSE Laboratory.

There are two possible routes:

Main Route:

College Gate $\rightarrow$ Block A $\rightarrow$ Library $\rightarrow$ Block B $\rightarrow$ CSE Lab.

Alternate Route:

College Gate $\rightarrow$ Canteen $\rightarrow$ Block B $\rightarrow$ CSE Lab.

Problem components.

Component

Initial state

Goal state.

State space.

Description.

College gate

CSE Laboratory

College gate, Block A, Library,
Canteen, Block B, CSE lab.

Path cost

total distance, time, or travel cost.

solution path

College gate $\rightarrow$ Block A $\rightarrow$ Library $\rightarrow$
Block B $\rightarrow$ CSE lab

Actions / operators.

1. move from college gate to Block A.
2. move from Block A to library.
3. move from library to Block B.
4. Move from canteen to Block B.
5. move from Block B to CSE lab.

State Space Representation

{ college gate, Block A, library, canteen, Block B, CSE lab }.

Possible solution 1

College gate $\rightarrow$ Block A $\rightarrow$ library $\rightarrow$ Block B $\rightarrow$
CSE lab.

Possible solution 2

College gate $\rightarrow$ Canteen $\rightarrow$ Block B $\rightarrow$ CSE lab

Conclusion

The college gate is the initial state & the CSE lab is the goal state. The AI searches through the available locations & selects a valid path.